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BITS Pilani
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AUTOTRONICS - AELZG533/AEZG533 -

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L2 & 3 – Electrical and electronics fundamentals

Outline

- Last Class
  - i. Introduction to the course
  - ii. Evaluation scheme
  - iii. Everything signal
- Today's class
  - i. Subsystem examples
  - ii. Fundamentals required
  - iii. Electrical and electronics in the vehicle

Clarifications

1) Difference betn linearity, hysteresis, accuracy & tolerance
↳ numerical

Sensors

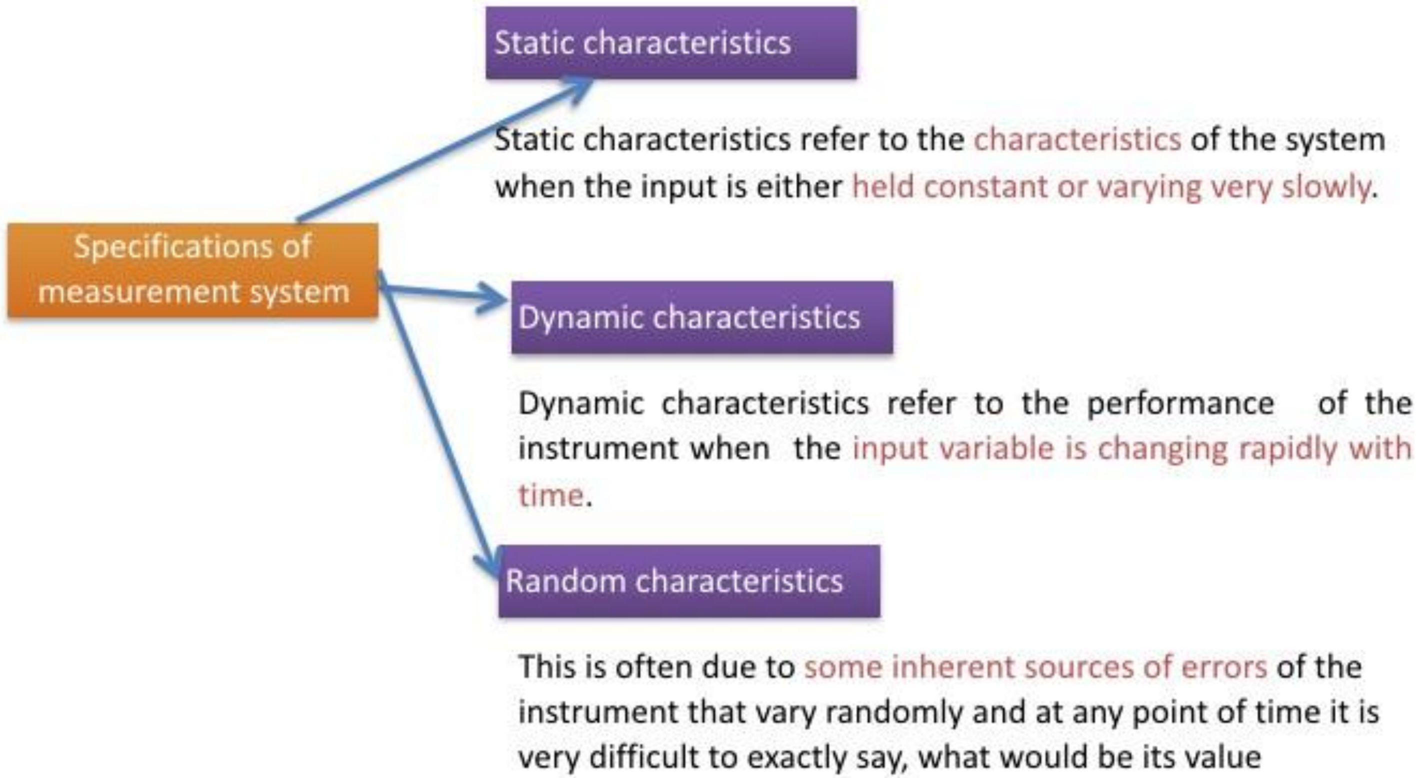
- The sensors and transducers aid in generating information to the output devices for control action.
- They participate in conversion of one domain of energy into another domain with or without the utilization of modulating domain.

Energy domains

- Mechanical
  - Distance
  - Velocity
  - Acceleration
  - Force
  - Etc.
- Electrical
  - Current
  - Resistance
  - Voltage
  - Inductance
  - Capacitance
- Magnetic
  - Field-strength
  - Flux density
- Thermal
  - Heat capacity
  - Latent heat
  - Phase changes
  - Sensible
- Radiant
  - Frequency
  - Phase
  - Intensity
  - Polarity
  - Etc.
- Chemical

Sensors - Characteristics

- Static and dynamic performance characteristics play important role in the selection of type of sensors needed.



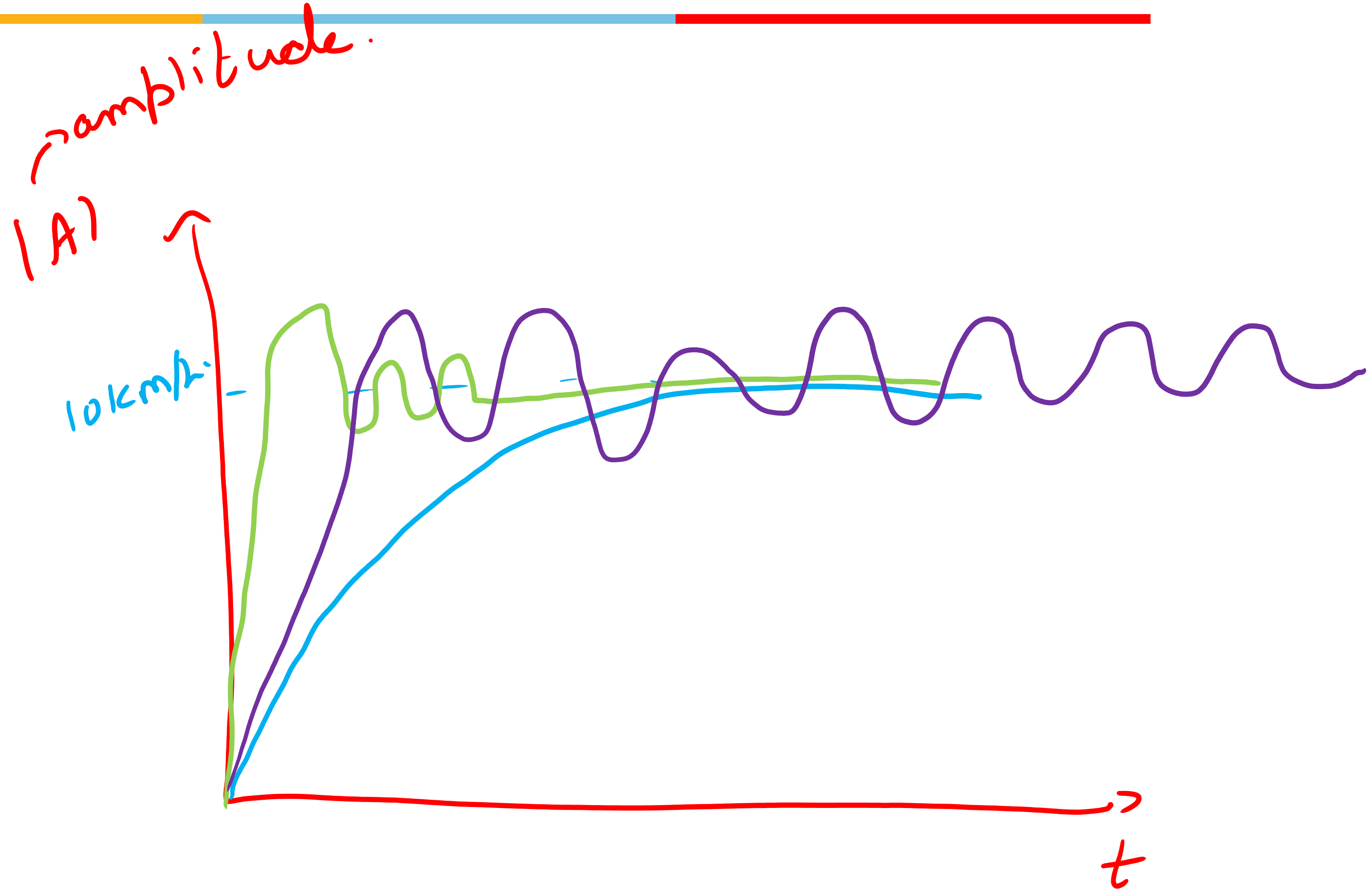
Sensors - Characteristics

Performance domain	Performance parameters
Static characteristics	• Range and span • Error • Accuracy • Repeatability • Resolution • Sensitivity • Stability • Impedance
Dynamic characteristics	• Response time • Setting time • Rise time • Time constants • Peak over shoot • Type of transient response

→ Propeties

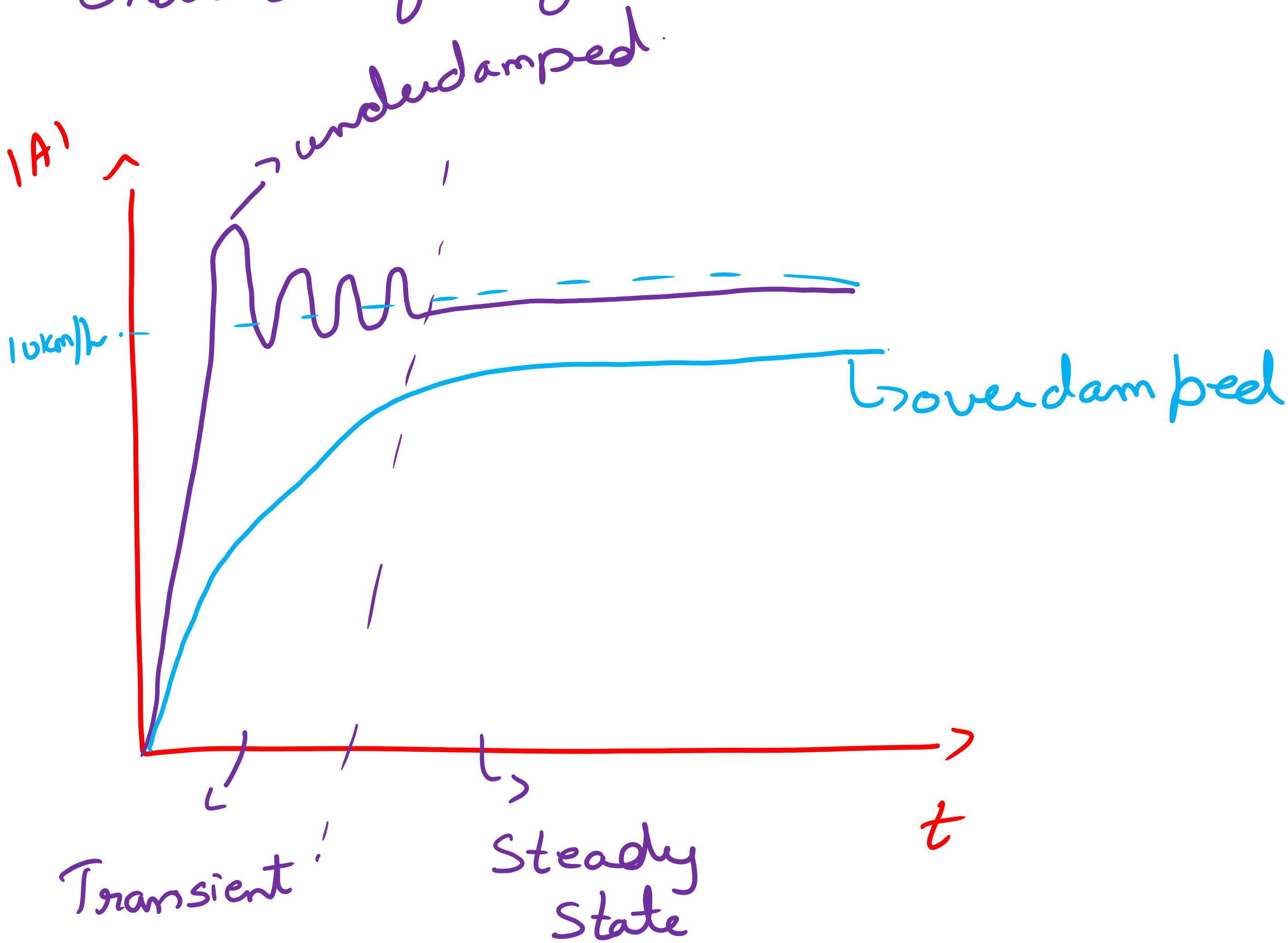
→ Transient

Types of responses



Types of responses

Charac. of a system.



Static characteristics

Range /span

It defines the maximum and minimum values of the inputs or the outputs for which the instrument is recommended to use. For example, for a temperature measuring instrument the input range may be 100-500 °C and the output range may be 4-20 mA.

(Example – Range – 100°C to 500°C, Span – 400°C)

Range – (max value – min value) → not subtraction.

LRV → low range value (LRV)

URV → upper range value (URV)

Span = (URV – LRV)

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Engine temp.

↳ temp. varies from 50°C to 100°C

↳ sensor range : 0°–150°C (Input)

(0–150mA) (output.

Span for measurement : 100 − 50 = 50°C

sensor span = 150 − 0 = 150°C

Sensitivity

Sensitivity

It can be defined as the ratio of the *incremental output* and the *incremental input* .
sensitivity of a thermocouple is denoted as 100µ*V*/°*C*.

Again sensitivity of an instrument may also vary with temperature or other external factors. This is known ***as sensitivity drift***.

Suppose the sensitivity of the spring balance mentioned above is 25 mm/kg at 20 ºC and 27 mm/kg at 30ºC. Then the sensitivity drift/ºC is 0.2 (mm/kg)/ºC.

temperature $0 - 100^\circ C$

voltage O P $0 - 4\,mV$.

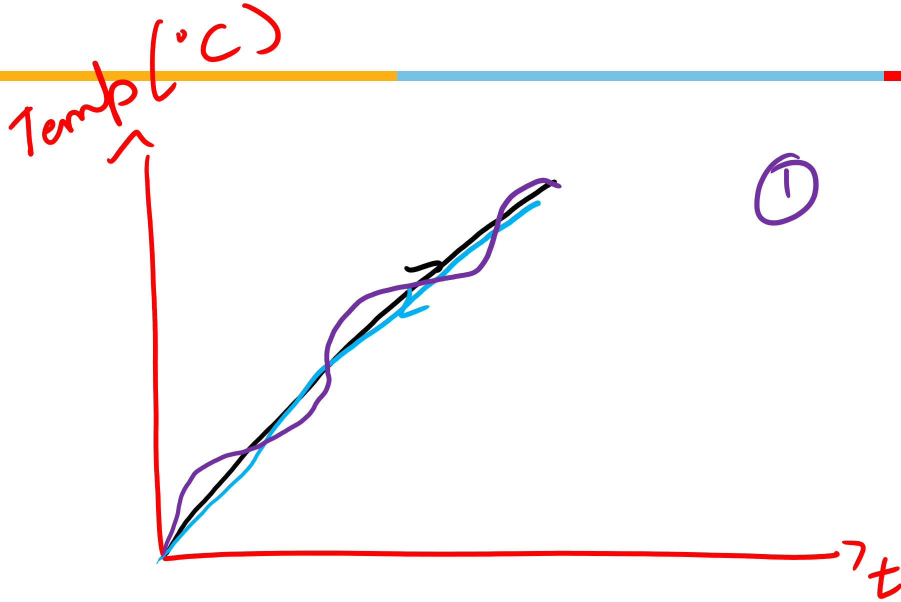
$$\text{Sensitivity} = \frac{\Delta OP}{\Delta IP} = \frac{4\,mV - 0}{100^\circ C - 0} = 0.04\,mV/^\circ C$$

temp.

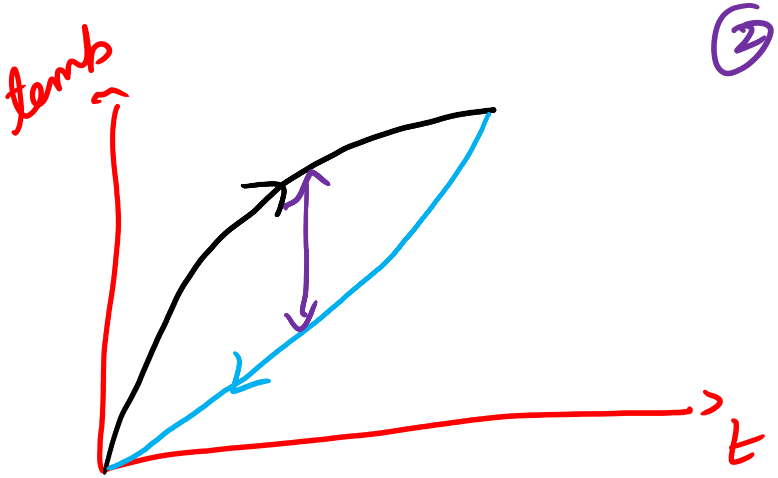
$20^\circ C \rightarrow 100\,\Omega$.

$120^\circ C \rightarrow 195\,\Omega$

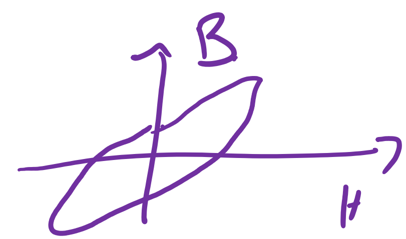
$$S = \frac{195 - 100}{120 - 20} = 0.95\,\Omega/^\circ C$$



① Linearity



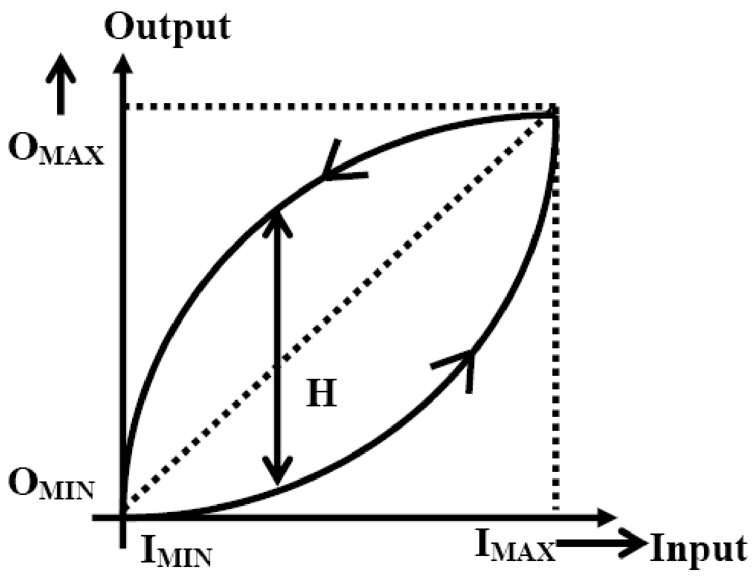
② Hysteresis



Sensors - Characteristics

- Hysteresis : Hysteresis exists not only in magnetic circuits, but in instruments also. The hysteresis is expressed as the maximum hysteresis as a full scale reading
  - For example, the deflection of a diaphragm type pressure gauge may be different for the same pressure, but one for increasing and other for decreasing, as shown in Fig.

$$Hysteresis = \frac{H}{O_{max} - O_{min}} X100.$$

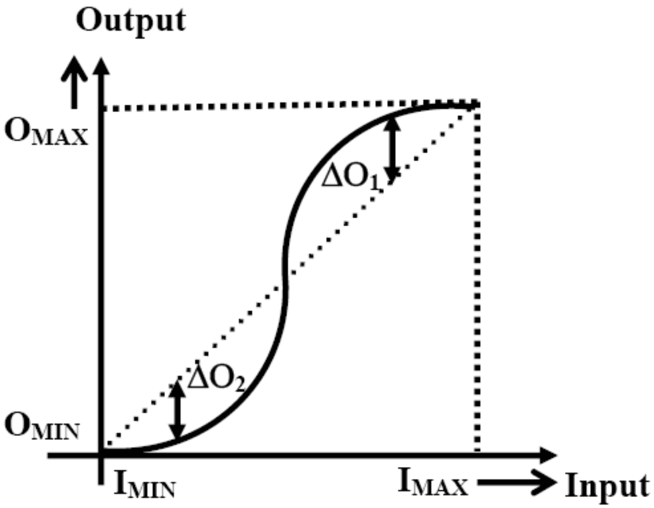


Sensors - Characteristics

- Linearity : The linearity is defined as the maximum deviation from the linear characteristics as a percentage of the full scale output.
  - Linearity is actually a measure of nonlinearity of the instrument.
  - When we talk about sensitivity, we assume that the input/output characteristic of the instrument to be approximately linear. But in practice, it is normally nonlinear, as shown in Fig.

$$Linearity = \frac{\Delta O}{O_{max} - O_{min}}$$

$$\Delta O = \max(\Delta O_1, \Delta O_2) .$$



Static characteristics

$$Linearity = \frac{\Delta O}{O_{max} - O_{min}} \qquad (1)$$

where, $\Delta O = \max(\Delta O_1, \Delta O_2)$.

$$Hysteresis = \frac{H}{O_{max} - O_{min}} X100.$$

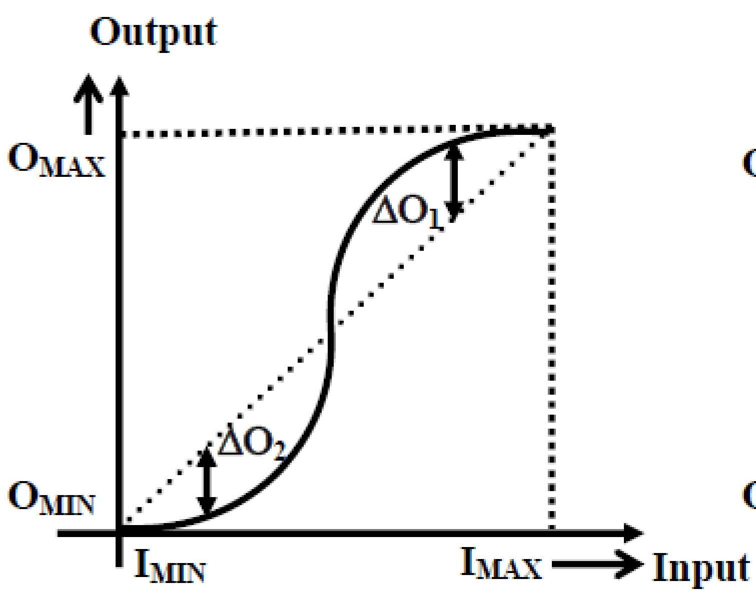


Fig. 1 Linearity

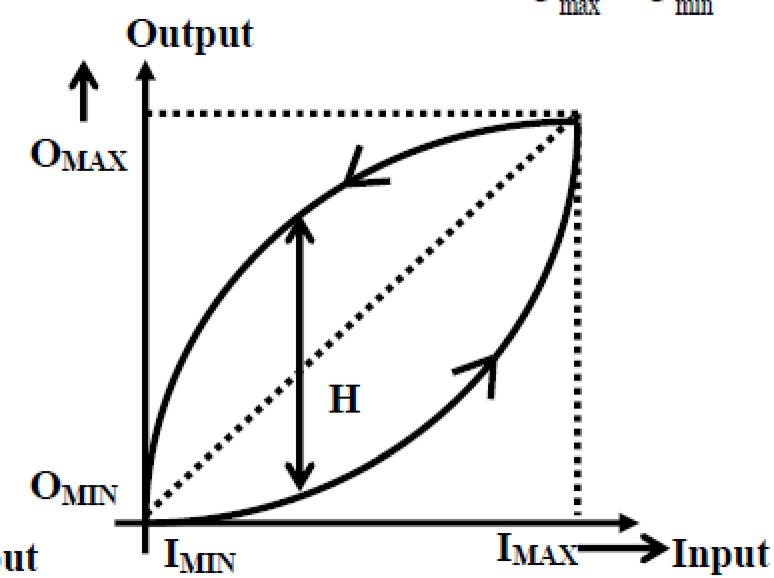


Fig. 2 Hysteresis

0 · S1 S2 S3 M1

0 · S1 S2 S3 S4 M2

0 · S1 S2 S3 S4 S5 M3

0.05 V

①

0.0537 V

② (0 - 20) V ①

(0 - 200) V ②

5V ideal

4.83V measured.

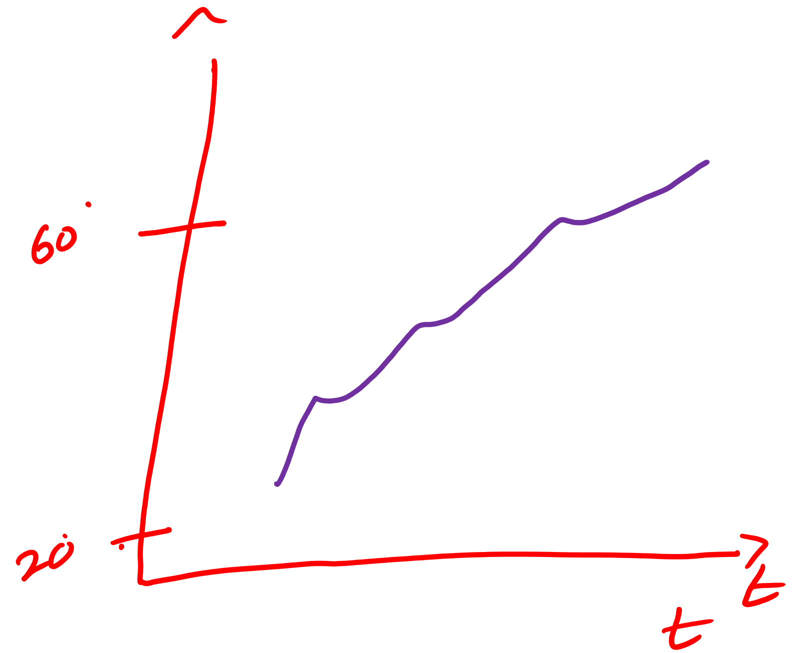
8% error

$$= \frac{5 - 4.83}{5} \times 100\%$$

Measuring 1V changes

Accuracy + resolution.


temp
100°c
50°
40°
0
t


60°
20°
t

Static characteristics

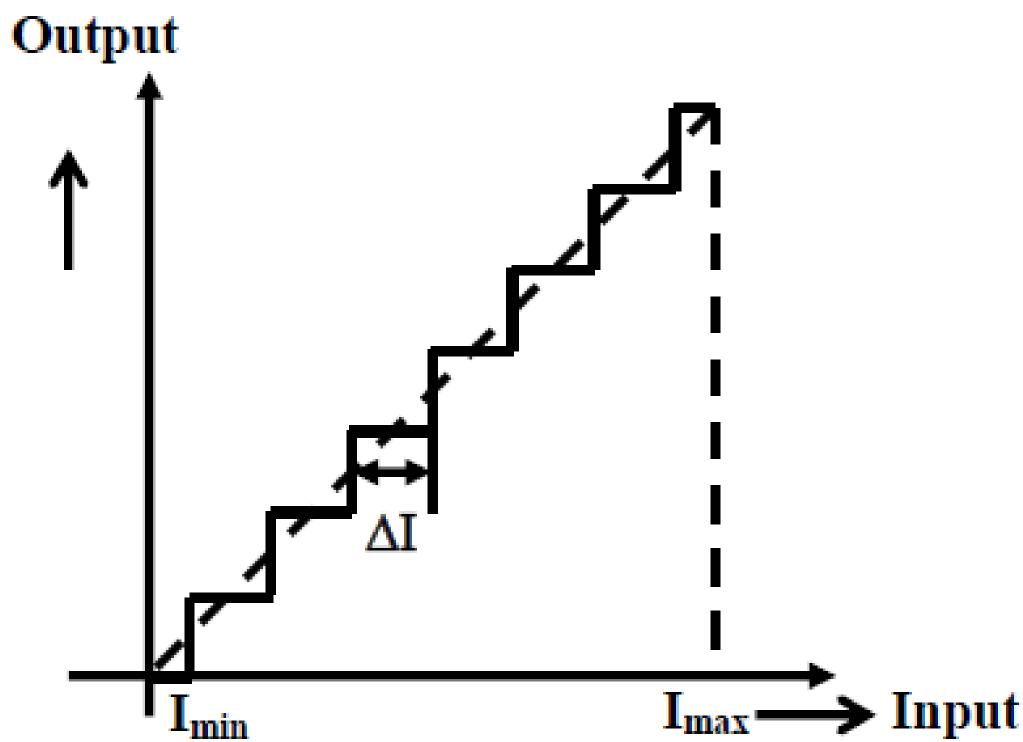


Fig. 3 Resolution

The smallest change in the input value that will produce an observable change in the output

Static characteristics

Accuracy

Accuracy indicates the closeness of the measured value with the actual or true value, and is expressed in the form of the *maximum error (= measured value – true value)* as a percentage of **full scale reading.**

↳ instrument

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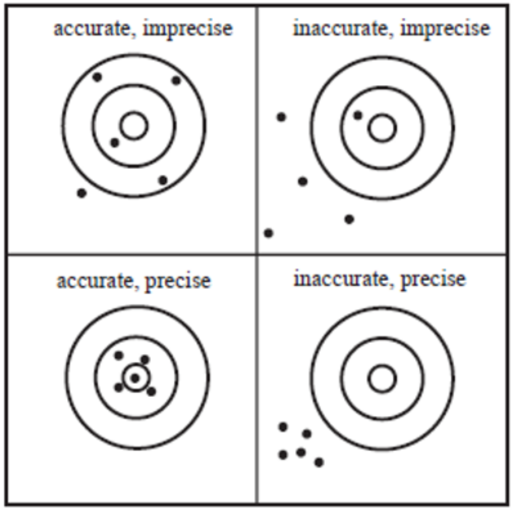
Precision

Precision indicates the repeatability or reproducibility of an instrument (but does not indicate accuracy).

$$Precision = \frac{measured\ range}{\sigma_e}$$



Static characteristics

Stability

Ability to give the same output when used to measure a constant input *over a period of time*

Dead band /time

Range of input values for which there is no output

(Inductive speed sensors produce no output initially at low RPMs)

Output Impedance

The value should be specified for Sensors giving electrical output interfaced with electronic circuits. By their very nature, they **impede** the measurement.

Sensors- Static Characteristics

- **Range and span:** It defi nes the maximum and minimum values of the inputs or the outputs for which the instrument is recommended to use.
  - For example, for a temperature measuring instrument the input range may be 100–500 degC and the output range may be 4–20 mA.
- **Error:** difference in true value and the measured value of the quantity being measured.
- **Accuracy:** Accuracy indicates the closeness of the measured value with the actual or true value, and is expressed in the form of the maximum error (= measured value – true value) as a percentage of full scale reading.
  - Thus, if the accuracy of a temperature indicator, with a full scale range of 0–500 degC is specified as ±0.5%, it indicates that the measured value will always be within ±2.5 degC of the true value.
- **Stability:** Ability to indicate the same output over a period of time for a constant i/p.
- **Impedance:** when the sensor connected in a electronic circuit, some form of disturbance (opposition that a circuit presents to the passage of a current) is known as impedance.

Sensors - Characteristics

- Repeatability: Ability to give same output reading when same i/p value is applied repeatedly. Precision indicates the repeatability or reproducibility of an instrument.

- A precision instrument indicates that the successive reading would be very close, or in other words, the standard deviation σe of the set of measurements would be very small. Quantitatively, the precision can be expressed as:

$$Precision = \frac{\overline{\text{measured range}}}{\sigma_e}$$

Precision means repetition of successive readings, but it does not guarantee accuracy; successive readings may be close to each other, but far from the true value.

Sensors - Characteristics

- Sensitivity: It can be defined as the ratio of the incremental output and the incremental input.

- We assume that the input–output characteristic of the instrument is approximately linear in that range.

  - Thus if the sensitivity of a thermocouple is denoted as 10µV/0C, it indicates the sensitivity in the linear range of the thermocouple voltage vs. temperature characteristics.

  - Similarly sensitivity of a spring balance can be expressed as 25 mm/kg (say), indicating additional load of 1 kg will cause additional displacement of the spring by 25mm.

Sensors - Characteristics

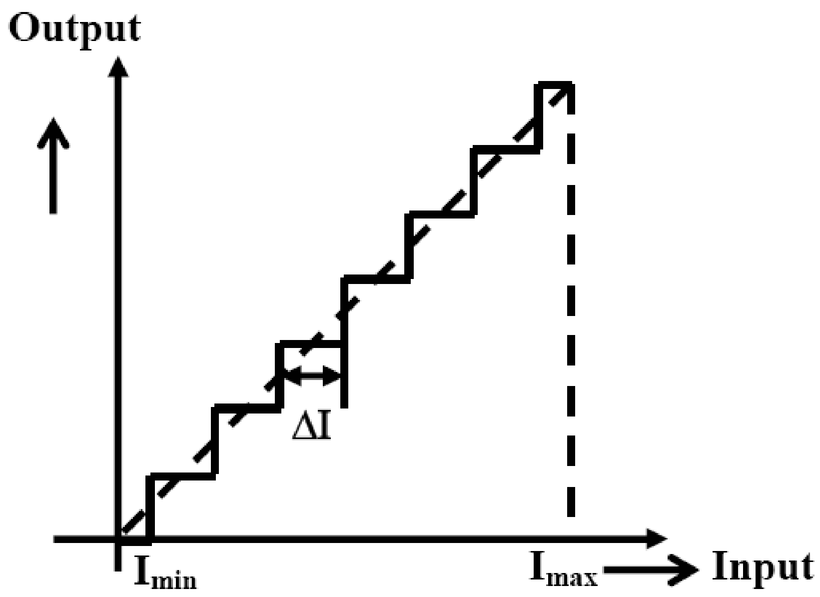
- Resolution : In some instruments, the output increases in discrete steps, for continuous increase in the input, as shown in Fig.

The smallest change in the input value that will produce an observable change in the output

For example, an eight-bit A/D converter with +5V input can measure the minimum voltage of

$$\frac{5}{2^8 - 1} \text{ or } 19.6\ mv$$

$$Resolution = \frac{\Delta I}{I_{max} - I_{min}} X100$$



Sensors- Dynamic characteristics

- Dynamic characteristics refer to the performance of the instrument when the input variable is changing rapidly with time.
- For example, human eye cannot detect any event whose duration is more than one-tenth of a second; thus the dynamic performance of human eye cannot be said to be very satisfactory.
- It is always convenient to express the input-output dynamic characteristics in form of a linear differential equation.

Sensors- Dynamic characteristics

- Dynamic characteristics specifications are normally referred to the performance of the instrument with different test signals, e.g. impulse input, step input, ramp input and sinusoidal input.

Step response performance

The normalized step response of a measurement system normally encountered is shown in Fig. Two important parameters for classifying the dynamic response are:

Peak Overshoot (M_p): It is the maximum value minus the steady state value, normally expressed in terms of percentage.

Settling Time (t_s): It is the time taken to attain the response within ±2% of the steady state value.

Rise time (t_r): It is the time required for the response to rise from 10% to 90% of its final value.

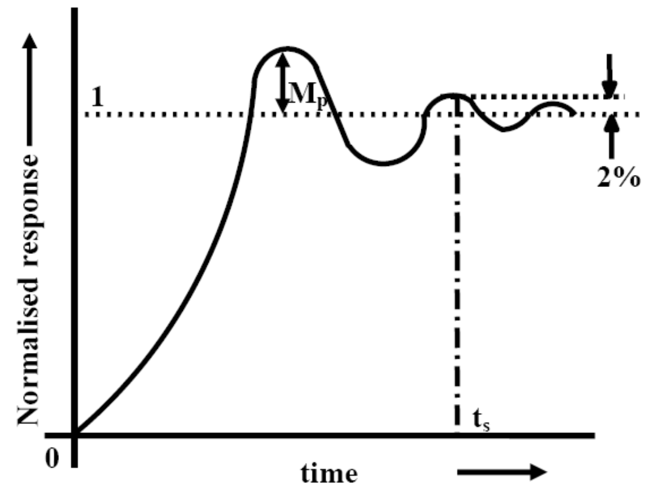


Fig. Step response of a dynamic system

Response time: time elapsed to give output signal from the point of supply of the input signal